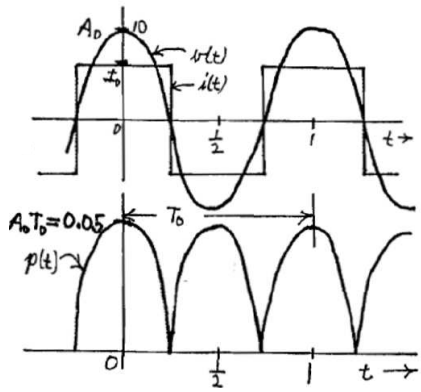
Assignment Chapter 2: Solution

Problem 2-3:
Part (a):

- The instantaneous power is given by:

$$p(t) = v(t)i(t)$$

- $v(t)$, $i(t)$ and $p(t)$ are plotted in the figure below:



- From the above figure, it can be observed that $p(t)$ can be written as:

$$p(t) = A_0 I_0 \left|\cos(\omega_0 t)\right| = 0.05 \left|\cos(2\pi t)\right|$$

Part (b):

- $p(t)$ is periodic with period $T_0/2 = 1/2$ implying that:

$$\begin{aligned}
\langle p(t) \rangle &= \frac{1}{1/2} \int_{-1/4}^{1/4} p(t) \mathrm{d}t \\
&= 0.1 \int_{-1/4}^{1/4} \cos(2\pi t) \mathrm{d}t \\
&= 0.1 \frac{1}{2\pi} \left[\sin\left(2\pi \frac{1}{4}\right) - \sin\left(-2\pi \frac{1}{4}\right) \right] \\
&= \frac{0.1}{2\pi} 2 \sin\left(\frac{\pi}{2}\right) = \frac{0.1}{\pi} = 0.0318 \qquad \text{(W)}
\end{aligned}$$

Problem 2-6:
Part (a):

- The energy of the signal is given by:

$$E = \int_{-\infty}^{+\infty} w^2(t)\mathrm{d}t = \int_{-T_0/2}^{+T_0/2} 1.\mathrm{d}t = T_0$$

  the signal has a finite nonzero energy implying that it is an energy signal.

Part (b):

- The energy of the signal is given by:

$$\begin{aligned} E &= \int_{-\infty}^{+\infty} w^2(t)\mathrm{d}t = \int_{-T_0/2}^{+T_0/2} \cos^2(\omega_0 t)\mathrm{d}t \\ &= \frac{1}{2}\int_{-T_0/2}^{+T_0/2} 1.\mathrm{d}t + \frac{1}{2}\underbrace{\int_{-T_0/2}^{+T_0/2} \cos(2\omega_0 t)\mathrm{d}t}_{=0} = \frac{T_0}{2} \end{aligned}$$

  implying that the signal is an energy signal.

Part (c):

- It can be directly observed that the energy of the signal is infinite since the area under the square of the signal is infinite.
- The average power of the signal is given by ($w(t) = \cos^2(\omega_0 t)$ is periodic with period $T_0/2$):

$$\begin{aligned} P &= \frac{2}{T_0}\int_{-T_0/4}^{+T_0/4} w^2(t)\mathrm{d}t = \frac{2}{T_0}\int_{-T_0/4}^{+T_0/4} \cos^4(\omega_0 t)\mathrm{d}t \\ &= \frac{2}{T_0}\int_{-T_0/4}^{+T_0/4} \left[\cos^2(\omega_0 t)\right]\left[\cos^2(\omega_0 t)\right]\mathrm{d}t \\ &= \frac{2}{T_0}\int_{-T_0/4}^{+T_0/4} \frac{1}{2}\left[1+\cos(2\omega_0 t)\right]\frac{1}{2}\left[1+\cos(2\omega_0 t)\right]\mathrm{d}t \\ &= \frac{2}{T_0}\frac{1}{4}\int_{-T_0/4}^{+T_0/4} \left[1+2\cos(2\omega_0 t)+\cos^2(2\omega_0 t)\right]\mathrm{d}t \\ &= \frac{1}{2T_0}\int_{-T_0/4}^{+T_0/4} \left[1+2\cos(2\omega_0 t)+\frac{1}{2}+\frac{1}{2}\cos(4\omega_0 t)\right]\mathrm{d}t \\ &= \frac{1}{2T_0}\left[\int_{-T_0/4}^{+T_0/4} \frac{3}{2}\mathrm{d}t + 2\underbrace{\int_{-T_0/4}^{+T_0/4} \cos(2\omega_0 t)\mathrm{d}t}_{=0} + \frac{1}{2}\underbrace{\int_{-T_0/4}^{+T_0/4} \cos(4\omega_0 t)\mathrm{d}t}_{=0}\right] = \frac{3}{8} \end{aligned}$$

  implying that P is finite and different from zero.

- Consequently, the signal is a power signal.

Problem 2-15:

- The Fourier transform of $w(t)$ is given by:

$$\begin{aligned} W(f) &= \mathcal{F}[w(t)] = \mathcal{F}[\sin(2\pi f_1 t)] * \mathcal{F}[\cos(2\pi f_2 t)] \\ &= \left[\frac{j}{2}\delta(f+f_1) - \frac{j}{2}\delta(f-f_1)\right] * \left[\frac{1}{2}\delta(f+f_2) + \frac{1}{2}\delta(f-f_2)\right] \end{aligned}$$

- On the other hand given that $x(f) * \delta(f+f_0) = x(f+f_0)$ then $W(f)$ can be written as:

$$W(f) = \frac{j}{4}\left[\delta(f+f_1+f_2) + \delta(f+f_1-f_2) - \delta(f-f_1+f_2) - \delta(f-f_1-f_2)\right]$$

Problem 2-17:

- The signal $w(t)$ can be written as:

$$w(t) = 4\text{rect}\left(\frac{t}{4}\right) - 2\text{rect}\left(\frac{t}{2}\right)$$

implying that:

$$w(t) = 16\text{sinc}\,(4f) - 4\text{sinc}\,(2f)$$

Problem 2-43:
Part (a):

- $R_w(\tau)$ is given by:

$$\begin{aligned} R_w(\tau) &= \langle w(t)w(t+\tau)\rangle \\ &= \langle [5+12\cos(\omega_0 t)][5+12\cos(\omega_0(t+\tau))]\rangle \\ &= \langle 25 + 60\cos(\omega_0 t) + 60\cos(\omega_0(t+\tau)) + 144\cos(\omega_0 t)\cos(\omega_0(t+\tau))\rangle \\ &= \underbrace{\langle 25\rangle}_{=25} + 60\underbrace{\langle\cos(\omega_0 t)\rangle}_{=0} + 60\underbrace{\langle\cos(\omega_0 t+\omega_0\tau)\rangle}_{=0} + \frac{1}{2}\left[144\underbrace{\langle\cos(2\omega_0 t+\omega_0\tau)\rangle}_{=0} + 144\underbrace{\langle\cos(\omega_0\tau)\rangle}_{=\cos(\omega_0\tau)}\right] \\ &= 25 + 72\cos(\omega_0\tau) \end{aligned}$$

- The PSD is given by:

$$\begin{aligned} \mathcal{P}_w(f) &= \mathcal{F}\left[R_w(\tau)\right] \\ &= \mathcal{F}\left[25 + 72\cos(2\pi f_0\tau)\right] \\ &= 25\delta(f) + 36\delta(f+f_0) + 36\delta(f-f_0) \end{aligned}$$

note that the Fourier transform was calculated with respect to the variable τ (and not t).

Problem 2-46:

- Define the integral I as:

$$I = \int_{-\frac{5}{2}T_2}^{\frac{5}{2}T_2} s_1(t)s_2^*(t)\mathrm{d}t = \int_{-\frac{5}{2}T_2}^{\frac{5}{2}T_2} s_1(t)s_2(t)\mathrm{d}t$$

  where the second equality follows since $s_2^*(t) = s_2(t)$ since the signal $s_2(t)$ is real.

- $s_1(t)$ and $s_2(t)$ are orthogonal over the interval $[-\frac{5}{2}T_2 \;\; \frac{5}{2}T_2]$ if and only if $I = 0$.

- I can be written as:

$$\begin{aligned} I &= A_1A_2 \int_{-\frac{5}{2}T_2}^{\frac{5}{2}T_2} \cos(\omega_1 t + \phi_1)\cos(\omega_2 t + \phi_2)\mathrm{d}t \\ &= \frac{A_1A_2}{2}\left[\underbrace{\int_{-\frac{5}{2}T_2}^{\frac{5}{2}T_2} \cos((\omega_1+\omega_2)t + \phi_1 + \phi_2)\mathrm{d}t}_{I_1} + \underbrace{\int_{-\frac{5}{2}T_2}^{\frac{5}{2}T_2} \cos((\omega_1-\omega_2)t + \phi_1 - \phi_2)\mathrm{d}t}_{I_2}\right] \end{aligned}$$

- I_1 can be written as:

$$I_1 = \begin{cases} 5T_2\cos(\phi_1+\phi_2), & \omega_1 = -\omega_2; \\ \frac{1}{\omega_1+\omega_2}\left[\sin\left((\omega_1+\omega_2)\frac{5\pi}{\omega_2} + \phi_1 + \phi_2\right) - \sin\left(-(\omega_1+\omega_2)\frac{5\pi}{\omega_2} + \phi_1 + \phi_2\right)\right], & \omega_1 \neq -\omega_2. \end{cases}$$

  however, given that $\omega_1 > 0$ and $\omega_2 > 0$ then ω_1 is always different from $-\omega_2$ implying that:

$$I_1 = \frac{1}{\omega_1+\omega_2}\left[\sin\left((\omega_1+\omega_2)\frac{5\pi}{\omega_2} + \phi_1 + \phi_2\right) - \sin\left(-(\omega_1+\omega_2)\frac{5\pi}{\omega_2} + \phi_1 + \phi_2\right)\right]$$

  for all values of ω_1, ω_2, ϕ_1 and ϕ_2.

- On the other hand the integral I_2 can be written as:

$$I_2 = \begin{cases} 5T_2\cos(\phi_1-\phi_2), & \omega_1 = \omega_2; \\ \frac{1}{\omega_1-\omega_2}\left[\sin\left((\omega_1-\omega_2)\frac{5\pi}{\omega_2} + \phi_1 - \phi_2\right) - \sin\left(-(\omega_1-\omega_2)\frac{5\pi}{\omega_2} + \phi_1 - \phi_2\right)\right], & \omega_1 \neq \omega_2. \end{cases}$$

Part (a):

- For $\omega_1 = \omega_2$ and $\phi_1 = \phi_2$:

$$\begin{aligned} I_1 &= \frac{1}{2\omega_2}\left[\sin\left((2\omega_2)\frac{5\pi}{\omega_2} + 2\phi_2\right) - \sin\left(-(2\omega_2)\frac{5\pi}{\omega_2} + 2\phi_2\right)\right] \\ &= \frac{1}{2\omega_2}\left[\sin(10\pi + 2\phi_2) - \sin(-10\pi + 2\phi_2)\right] \\ &= \frac{1}{2\omega_2}\left[\sin(2\phi_2) - \sin(2\phi_2)\right] = 0 \\ I_2 &= 5T_2\cos(0) = 5T_2 \end{aligned}$$

  resulting in $I = \frac{5A_1A_2T_2}{2} \neq 0$ implying that $s_1(t)$ and $s_2(t)$ are not orthogonal.

Part (b):

- For $\omega_1 = \omega_2$ and $\phi_1 = \phi_2 + \pi/2$:

$$\begin{aligned}
I_1 &= \frac{1}{2\omega_2}\left[\sin\left((2\omega_2)\frac{5\pi}{\omega_2} + 2\phi_2 + \pi/2\right) - \sin\left(-(2\omega_2)\frac{5\pi}{\omega_2} + 2\phi_2 + \pi/2\right)\right] \\
&= \frac{1}{2\omega_2}\left[\sin(10\pi + 2\phi_2 + \pi/2) - \sin(-10\pi + 2\phi_2 + \pi/2)\right] \\
&= \frac{1}{2\omega_2}\left[\cos(2\phi_2) - \cos(2\phi_2)\right] = 0 \\
I_2 &= 5T_2\cos(\pi/2) = 0
\end{aligned}$$

resulting in $I = 0$ implying that $s_1(t)$ and $s_2(t)$ are orthogonal.

Part (c):

- For $\omega_1 = \omega_2$ and $\phi_1 = \phi_2 + \pi$:

$$\begin{aligned}
I_1 &= \frac{1}{2\omega_2}\left[\sin\left((2\omega_2)\frac{5\pi}{\omega_2} + 2\phi_2 + \pi\right) - \sin\left(-(2\omega_2)\frac{5\pi}{\omega_2} + 2\phi_2 + \pi\right)\right] \\
&= \frac{1}{2\omega_2}\left[\sin(10\pi + 2\phi_2 + \pi) - \sin(-10\pi + 2\phi_2 + \pi)\right] \\
&= \frac{1}{2\omega_2}\left[-\sin(2\phi_2) + \sin(2\phi_2)\right] = 0 \\
I_2 &= 5T_2\cos(\pi) = -5T_2
\end{aligned}$$

resulting in $I = -\frac{5A_1A_2T_2}{2} \neq 0$ implying that $s_1(t)$ and $s_2(t)$ are not orthogonal.

Part (d):

- For $\omega_1 = 2\omega_2$ and $\phi_1 = \phi_2$:

$$\begin{aligned}
I_1 &= \frac{1}{3\omega_2}\left[\sin\left((3\omega_2)\frac{5\pi}{\omega_2} + 2\phi_2\right) - \sin\left(-(3\omega_2)\frac{5\pi}{\omega_2} + 2\phi_2\right)\right] \\
&= \frac{1}{3\omega_2}\left[\sin(15\pi + 2\phi_2) - \sin(-15\pi + 2\phi_2)\right] \\
&= \frac{1}{3\omega_2}\left[-\sin(2\phi_2) + \sin(2\phi_2)\right] = 0 \\
I_2 &= \frac{1}{\omega_2}\left[\sin\left((\omega_2)\frac{5\pi}{\omega_2}\right) - \sin\left(-(\omega_2)\frac{5\pi}{\omega_2}\right)\right] \\
&= \frac{1}{\omega_2}\left[\sin(5\pi) + \sin(5\pi)\right] = \frac{2\sin(5\pi)}{\omega_2} = 0
\end{aligned}$$

resulting in $I = 0$ implying that $s_1(t)$ and $s_2(t)$ are orthogonal.

Part (e):

- For $\omega_1 = \frac{4}{5}\omega_2$ and $\phi_1 = \phi_2$:

$$
\begin{aligned}
I_1 &= \frac{5}{9\omega_2}\left[\sin\left((\frac{9}{5}\omega_2)\frac{5\pi}{\omega_2} + 2\phi_2\right) - \sin\left(-(\frac{9}{5}\omega_2)\frac{5\pi}{\omega_2} + 2\phi_2\right)\right] \\
&= \frac{5}{9\omega_2}\left[\sin(9\pi + 2\phi_2) - \sin(-9\pi + 2\phi_2)\right] \\
&= \frac{5}{9\omega_2}\left[-\sin(2\phi_2) + \sin(2\phi_2)\right] = 0 \\
I_2 &= \frac{5}{\omega_2}\left[\sin\left((\frac{1}{5}\omega_2)\frac{5\pi}{\omega_2}\right) - \sin\left(-(\frac{1}{5}\omega_2)\frac{5\pi}{\omega_2}\right)\right] \\
&= \frac{5}{\omega_2}\left[\sin(\pi) - \sin(-\pi)\right] = \frac{5}{\omega_2}2\sin(\pi) = 0
\end{aligned}
$$

resulting in $I = 0$ implying that $s_1(t)$ and $s_2(t)$ are orthogonal.

Part (f):

- For $\omega_1 = \pi\omega_2$ and $\phi_1 = \phi_2$:

$$
\begin{aligned}
I_1 &= \frac{1}{(1+\pi)\omega_2}\left[\sin\left(((1+\pi)\omega_2)\frac{5\pi}{\omega_2} + 2\phi_2\right) - \sin\left(-((1+\pi)\omega_2)\frac{5\pi}{\omega_2} + 2\phi_2\right)\right] \\
&= \frac{1}{(1+\pi)\omega_2}\left[\sin(5\pi + 5\pi^2 + 2\phi_2) - \sin(-5\pi - 5\pi^2 + 2\phi_2)\right] \\
&= \frac{1}{(1+\pi)\omega_2}\left[-\sin(5\pi^2 + 2\phi_2) + \sin(-5\pi^2 + 2\phi_2)\right] \neq 0 \\
I_2 &= \frac{1}{(1-\pi)\omega_2}\left[\sin\left(((1-\pi)\omega_2)\frac{5\pi}{\omega_2}\right) - \sin\left(-((1-\pi)\omega_2)\frac{5\pi}{\omega_2}\right)\right] \\
&= \frac{1}{(1-\pi)\omega_2}\left[\sin\left(5\pi - 5\pi^2\right) - \sin\left(-5\pi + 5\pi^2\right)\right] \\
&= \frac{1}{(1-\pi)\omega_2}\left[\sin\left(5\pi^2\right) + \sin\left(5\pi^2\right)\right] \neq 0
\end{aligned}
$$

consequently, $I \neq 0$ implying that $s_1(t)$ and $s_2(t)$ are not orthogonal.

Problem 2-49:
Part (a):

- Consider the functions $\phi_1(t)$ and $\phi_2(t)$:

$$\begin{aligned} I_{12} &= \int_{-4}^{4} \phi_1(t)\phi_2^*(t)\mathrm{d}t = \int_{-4}^{4} \phi_1(t)\phi_2(t)\mathrm{d}t \\ &= \int_{-4}^{0} (1)(1)\mathrm{d}t + \int_{0}^{4} (1)(-1)\mathrm{d}t = 4 - 4 = 0 \end{aligned}$$

- Consider the functions $\phi_1(t)$ and $\phi_3(t)$:

$$\begin{aligned} I_{13} &= \int_{-4}^{4} \phi_1(t)\phi_3^*(t)\mathrm{d}t = \int_{-4}^{4} \phi_1(t)\phi_3(t)\mathrm{d}t \\ &= \int_{-4}^{-2} (1)(-1)\mathrm{d}t + \int_{-2}^{2} (1)(1)\mathrm{d}t + \int_{2}^{4} (1)(-1)\mathrm{d}t = -2 + 4 - 2 = 0 \end{aligned}$$

- Consider the functions $\phi_2(t)$ and $\phi_3(t)$:

$$\begin{aligned} I_{23} &= \int_{-4}^{4} \phi_2(t)\phi_3^*(t)\mathrm{d}t = \int_{-4}^{4} \phi_2(t)\phi_3(t)\mathrm{d}t \\ &= \int_{-4}^{-2} (1)(-1)\mathrm{d}t + \int_{-2}^{0} (1)(1)\mathrm{d}t + \int_{0}^{2} (-1)(1)\mathrm{d}t + \int_{2}^{4} (-1)(-1)\mathrm{d}t \\ &= -2 + 2 - 2 + 2 = 0 \end{aligned}$$

- Consequently, $I_{12} = I_{13} = I_{23} = 0$ implying that the functions in the set $\{\phi_i(t)\}_{i=1}^{3}$ are orthogonal over the interval $[-4 \;\; 4]$.

Part (b):

- Consider the function $\phi_1(t)$:

$$\begin{aligned} I_1 &= \int_{-4}^{4} \phi_1(t)\phi_1^*(t)\mathrm{d}t = \int_{-4}^{4} \phi_1^2(t)\mathrm{d}t \\ &= \int_{-4}^{4} (1)^2\mathrm{d}t = 8 \end{aligned}$$

- Consider the function $\phi_2(t)$:

$$\begin{aligned} I_2 &= \int_{-4}^{4} \phi_2(t)\phi_2^*(t)\mathrm{d}t = \int_{-4}^{4} \phi_2^2(t)\mathrm{d}t \\ &= \int_{-4}^{0} (1)^2\mathrm{d}t + \int_{0}^{4} (-1)^2\mathrm{d}t = 4 + 4 = 8 \end{aligned}$$

- Consider the function $\phi_3(t)$:

$$\begin{aligned} I_3 &= \int_{-4}^{4} \phi_3(t)\phi_3^*(t)\mathrm{d}t = \int_{-4}^{4} \phi_3^2(t)\mathrm{d}t \\ &= \int_{-4}^{-2} (-1)^2\mathrm{d}t + \int_{-2}^{2} (1)^2\mathrm{d}t + \int_{2}^{4} (-1)^2\mathrm{d}t = 2 + 4 + 2 = 8 \end{aligned}$$

- Define the functions $\{\phi_i'(t)\}_{i=1}^3$ as:

$$\phi_1'(t) = \frac{1}{\sqrt{I_1}}\phi_1(t) = \frac{1}{\sqrt{8}}\phi_1(t)$$
$$\phi_2'(t) = \frac{1}{\sqrt{I_2}}\phi_2(t) = \frac{1}{\sqrt{8}}\phi_2(t)$$
$$\phi_3'(t) = \frac{1}{\sqrt{I_3}}\phi_3(t) = \frac{1}{\sqrt{8}}\phi_3(t)$$

  The functions in the set $\{\phi_i'(t)\}_{i=1}^3 = \{\frac{1}{\sqrt{8}}\phi_i(t)\}_{i=1}^3$ are orthonormal.

Part (c):

- $w(t)$ can be written as:

$$w(t) = \sum_{i=1}^{3} a_i\phi_i(t) = a_1\phi_1(t) + a_2\phi_2(t) + a_3\phi_3(t)$$

  where:

$$\begin{aligned} a_i &= \frac{1}{I_i}\int_{-4}^{4} w(t)\phi_i^*(t)\mathrm{d}t = \frac{1}{8}\int_{-4}^{4} w(t)\phi_i(t)\mathrm{d}t \\ &= \frac{1}{8}\int_{0}^{4} \phi_i(t)\mathrm{d}t \end{aligned}$$

  where the last equality follows since $w(t) = 1$ for $0 \leq t \leq 4$ and $w(t) = 0$ elsewhere.

- Consequently:

$$a_1 = \frac{1}{8}\int_0^4 \phi_1(t)\mathrm{d}t = \frac{1}{8}(4) = \frac{1}{2}$$
$$a_2 = \frac{1}{8}\int_0^4 \phi_2(t)\mathrm{d}t = \frac{1}{8}(-4) = -\frac{1}{2}$$
$$a_3 = \frac{1}{8}\int_0^4 \phi_3(t)\mathrm{d}t = \frac{1}{8}(2-2) = 0$$

- Therefore:

$$\begin{aligned} w(t) &= \frac{1}{2}\left[\phi_1(t) - \phi_2(t)\right] \\ &= \frac{1}{2}\left[\sqrt{8}\phi_1'(t) - \sqrt{8}\phi_2'(t)\right] \\ &= \sqrt{2}\phi_1'(t) - \sqrt{2}\phi_2'(t) \end{aligned}$$

Part (d):

- From part (c), $w(t)$ can be written as:

$$\begin{aligned} w(t) &= \frac{1}{2}\left[\phi_1(t) - \phi_2(t)\right] \\ &= \sum_{j=1}^{3} a_j\phi_j(t) = a_1\phi_1(t) + a_2\phi_2(t) + a_3\phi_3(t) \end{aligned}$$

  where $a_1 = \frac{1}{2}$, $a_2 = -\frac{1}{2}$ and $a_3 = 0$.

- Consequently:

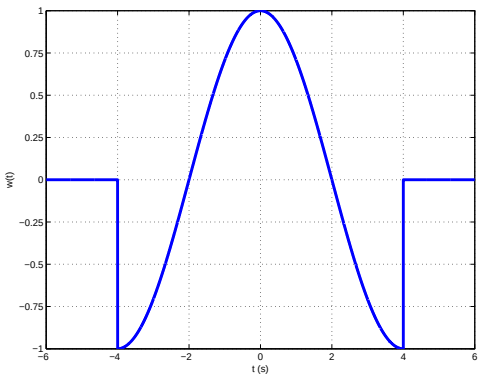
$$w(t) - \sum_{j=1}^{3} a_j \phi_j(t) = 0$$

resulting in a mean square error of:

$$\epsilon = \int_{-4}^{4} \left[w(t) - \sum_{j=1}^{3} a_j \phi_j(t) \right]^2 \mathrm{d}t = \int_{-4}^{4} [0]^2 \, \mathrm{d}t = 0$$

Part (e):

- The signal $w(t)$ is plotted in the figure below:



- The coefficient a_1 is given by:

$$a_1 = \frac{1}{8} \int_{-4}^{+4} w(t)\phi_1(t)\mathrm{d}t = \frac{1}{8} \int_{-4}^{+4} w(t)\mathrm{d}t$$

where the second equality follows since $\phi_1(t) = 1$ for $-4 \leq t \leq 4$. Consequently:

$$a_1 = \frac{1}{8} \int_{-4}^{+4} \cos\left(\frac{2\pi}{8} t\right) \mathrm{d}t = 0$$

since $\cos\left(\frac{2\pi}{8} t\right)$ has a period of 8 and the above integral is over a duration of 1 period $(4 - (-4) = 8)$.

- The coefficient a_2 is given by:

$$a_2 = \frac{1}{8} \int_{-4}^{+4} w(t)\phi_2(t)\mathrm{d}t = 0$$

since $w(t)$ is even and $\phi_2(t)$ is odd implying that $w(t)\phi_2(t)$ is odd and $\int_{-a}^{+a} f(t)\mathrm{d}t = 0$ when $f(t)$ is odd.

- The coefficient a_3 is given by:

$$a_3 = \frac{1}{8} \int_{-4}^{+4} w(t)\phi_3(t)\mathrm{d}t = \frac{1}{8} 2 \int_{0}^{+4} w(t)\phi_3(t)\mathrm{d}t$$

since $w(t)$ is even and $\phi_3(t)$ is even implying that $w(t)\phi_3(t)$ is even and $\int_{-a}^{a} f(t)\mathrm{d}t = 2\int_0^a f(t)\mathrm{d}t$ when $f(t)$ is even. Consequently:

$$\begin{aligned}
a_3 &= \frac{1}{4}\left[\int_0^2 \cos\left(\frac{2\pi}{8}t\right)\mathrm{d}t - \int_2^4 \cos\left(\frac{2\pi}{8}t\right)\mathrm{d}t\right] \\
&= \frac{1}{4}\left[\frac{8}{2\pi}\left[\sin\left(\frac{2\pi}{8}2\right) - \sin\left(\frac{2\pi}{8}0\right)\right] - \frac{8}{2\pi}\left[\sin\left(\frac{2\pi}{8}4\right) - \sin\left(\frac{2\pi}{8}2\right)\right]\right] \\
&= \frac{1}{\pi}\left[[1-0]-[0-1]\right] = \frac{2}{\pi}
\end{aligned}$$

- The mean square error can be written as:

$$\begin{aligned}
\epsilon &= \int_{-4}^{+4}\left[w(t) - \sum_{j=1}^{3} a_j\phi_j(t)\right]^2 \mathrm{d}t = \int_{-4}^{+4}\left[w(t) - \frac{2}{\pi}\phi_3(t)\right]^2 \mathrm{d}t \\
&= \underbrace{\int_{-4}^{+4} w^2(t)\mathrm{d}t}_{=I_1} - \frac{4}{\pi}\underbrace{\int_{-4}^{+4} w(t)\phi_3(t)\mathrm{d}t}_{=I_2} + \frac{4}{\pi^2}\underbrace{\int_{-4}^{+4}\phi_3^2(t)\mathrm{d}t}_{=I_3}
\end{aligned}$$

  - I_1 can be written as:

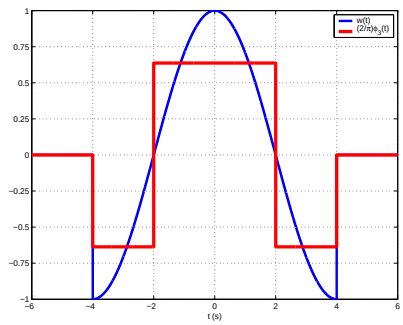
$$I_1 = 8\frac{1}{8}\int_{-4}^{+4} w^2(t)\mathrm{d}t = 8\left\langle \cos^2\left(\frac{2\pi}{8}t\right)\right\rangle = 8\frac{1}{2} = 4$$

    since the power of the sinusoidal function $\cos\left(\frac{2\pi}{8}t\right)$ is $\left\langle\cos^2\left(\frac{2\pi}{8}t\right)\right\rangle = \frac{1}{2}$.
  - On the other hand, $I_2 = 8a_3 = \frac{16}{\pi}$.
  - Finally, $I_3 = \int_{-4}^{+4} 1.\mathrm{d}t = 8$.

- Therefore:

$$\epsilon = 4 - \frac{64}{\pi^2} + \frac{32}{\pi^2} = 4 - \frac{32}{\pi^2} = 0.7577$$

  Since $\epsilon \neq 0$, the set $\{\phi_i(t)\}_{i=1}^{3}$ does not form a complete orthogonal set. In other words, the equality $w(t) = \sum_{i=1}^{3} a_i\phi_i(t) = \frac{2}{\pi}\phi_3(t)$ is an approximation. The functions $w(t)$ and $\frac{2}{\pi}\phi_3(t)$ are plotted in the figure below:



Problem 2-74:
Part (a):

- $x(t)$ can be written as:

$$x(t) = 0.5 + 1.5\cos\left(2\pi\frac{1}{3}t\right) + 0.5\sin\left(2\pi\frac{1}{3}t\right)$$

  implying that its Fourier transform can be written as:

$$\begin{aligned} X(f) &= 0.5\delta(f) + 0.75\delta\left(f+\frac{1}{3}\right) + 0.75\delta\left(f-\frac{1}{3}\right) + 0.25j\delta\left(f+\frac{1}{3}\right) - 0.25j\delta\left(f-\frac{1}{3}\right) \\ &= 0.5\delta(f) + (0.75+0.25j)\delta\left(f+\frac{1}{3}\right) + (0.75-0.25j)\delta\left(f-\frac{1}{3}\right) \end{aligned}$$

- Note that the fundamental frequency of $x(t)$ is $f_0 = 1/3$ implying that $X(f)$ can be written as:

$$X(f) = \sum_n c_n \delta(f - nf_0)$$

  where $c_0 = 0.5$, $c_{-1} = 0.75 + 0.25j$, $c_1 = 0.75 - 0.25j$ and $c_n = 0\ ;\ n \neq 0, \pm 1$.

- Consequently, the input PSD is given by:

$$\begin{aligned} \mathcal{P}_x(f) &= \sum_n |c_n|^2 \delta(f - nf_0) \\ &= 0.25\delta(f) + 0.625\delta\left(f+\frac{1}{3}\right) + 0.625\delta\left(f-\frac{1}{3}\right) \end{aligned}$$

Part (b):

- The transfer function of the filter is given by:

$$H(f) = \frac{1}{1 + j(2\pi RC)f} = \frac{1}{1 + j2\pi f} \quad \Rightarrow \quad |H(f)|^2 = \frac{1}{1 + 4\pi^2 f^2}$$

- The output PSD is given by:

$$\begin{aligned} \mathcal{P}_y(f) &= |H(f)|^2 \mathcal{P}_x(f) \\ &= \left[\frac{1}{1+4\pi^2 f^2}\right]_{f=0} [0.25\delta(f)] + \left[\frac{1}{1+4\pi^2 f^2}\right]_{f=-\frac{1}{3}} \left[0.625\delta\left(f+\frac{1}{3}\right)\right] \\ &\qquad + \left[\frac{1}{1+4\pi^2 f^2}\right]_{f=\frac{1}{3}} \left[0.625\delta\left(f-\frac{1}{3}\right)\right] \\ &= 0.25\delta(f) + \frac{0.625}{1+4\pi^2\left(\frac{1}{3}\right)^2}\left[\delta\left(f+\frac{1}{3}\right) + \delta\left(f-\frac{1}{3}\right)\right] \\ &= 0.25\delta(f) + 0.116\delta\left(f+\frac{1}{3}\right) + 0.116\delta\left(f-\frac{1}{3}\right) \end{aligned}$$

Part (c):

- P_y is given by:

$$
\begin{aligned}
P_y &= \int_{-\infty}^{+\infty} \mathcal{P}_y(f)\mathrm{d}f \\
&= 0.25 \underbrace{\int_{-\infty}^{+\infty} \delta(f)\mathrm{d}f}_{=1} + 0.116 \underbrace{\int_{-\infty}^{+\infty} \delta\left(f+\frac{1}{3}\right)\mathrm{d}f}_{=1} + 0.116 \underbrace{\int_{-\infty}^{+\infty} \delta\left(f-\frac{1}{3}\right)\mathrm{d}f}_{=1} \\
&= 0.25 + 0.116 + 0.116 = 0.482 \qquad \text{(W)}
\end{aligned}
$$

Problem 2-77:
Part (a):

- The relation between the output and the input is given by:

$$y(t) = x(t) - x(t - T_d)$$

implying that:

$$Y(f) = X(f) - X(f)e^{-j2\pi T_d f}$$

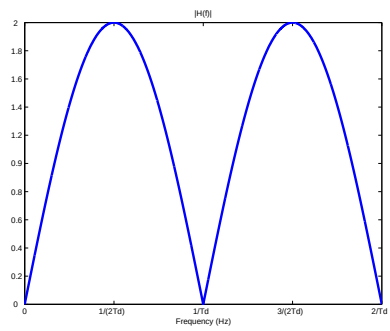
- Consequently, the transfer function is given by:

$$H(f) = \frac{Y(f)}{X(f)} = 1 - e^{-j2\pi T_d f}$$

- $H(f)$ can be written as:

$$
\begin{aligned}
H(f) &= e^{-j\pi T_d f}\left[e^{j\pi T_d f} - e^{-j\pi T_d f}\right] \\
&= 2je^{-j\pi T_d f}\sin\left(\pi T_d f\right) \\
\Rightarrow \quad |H(f)| &= 2\left|\sin\left(\pi T_d f\right)\right|
\end{aligned}
$$

- $|H(f)|$ is shown in the figure below:



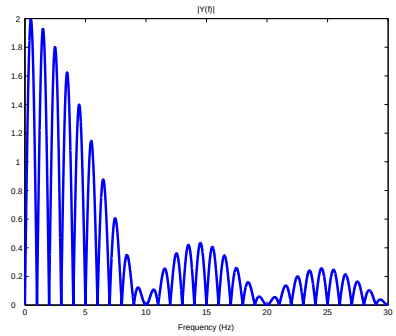
note that $|H(f)|$ is periodic with period $1/T_d$.

Part (b):

- For $x(t) = \text{rect}(t)$, $X(f) = \text{sinc}(f)$ implying that:

$$\begin{aligned} & Y(f) = H(f)X(f) \\ \Rightarrow \quad & |Y(f)| = |H(f)||X(f)| = 2\,|\sin(\pi T_d f)|\,|\text{sinc}(f)| \end{aligned}$$

- Using a computer, $|Y(f)|$ is plotted in the figure below for $T_d = 0.1$:



Problem 2-82:
Part (a):

- According to the sampling theorem:

$$f_s \geq 2f_m = 2\left(\frac{500}{2\pi}\right) = \frac{500}{\pi} = 159.15 \qquad \text{(Hz)}$$

implying that:

$$T_s = \frac{1}{f_s} \leq \frac{\pi}{500} = 6.283 \qquad \text{(ms)}$$

Part (b):

- The number of samples is given by:

$$\begin{aligned} N &= f_s t \\ &= [159.15 \ \ \text{samples/s}]\,[1 \ \ \text{s}] \simeq 159 \qquad \text{(samples)} \end{aligned}$$

Problem 2-89:

- The transfer function of the filter is given by:

$$H(f) = \frac{1}{1 + j(2\pi RC)f} \quad \Rightarrow \quad |H(f)|^2 = \frac{1}{1 + 4\pi^2(RC)^2 f^2}$$

- The 3-dB bandwidth B_{3dB} can be determined from:

$$\begin{aligned} & |H(B_{3dB})|^2 = \frac{1}{2}\left[\max|H(f)|^2\right] = \frac{1}{2}\left[|H(f)|^2\right]_{f=0} = \frac{1}{2} \\ \Rightarrow \quad & \frac{1}{1 + 4\pi^2(RC)^2 B_{3dB}^2} = \frac{1}{2} \\ \Rightarrow \quad & 4\pi^2(RC)^2 B_{3dB}^2 = 1 \quad \Rightarrow \quad B_{3dB} = \frac{1}{2\pi RC} \end{aligned}$$

Part (a):

- The equivalent bandwidth is given by:

$$B_{eq} = \frac{1}{|H(f_0)|^2}\int_0^{+\infty} |H(f)|^2 \mathrm{d}f$$
$$= \frac{1}{[|H(f)|^2]_{f=0}}\int_0^{+\infty} \frac{1}{1+4\pi^2(RC)^2f^2}\mathrm{d}f = \int_0^{+\infty} \frac{1}{1+4\pi^2(RC)^2f^2}\mathrm{d}f$$

Let $F = 2\pi RCf$, then $\mathrm{d}F = 2\pi RC\mathrm{d}f$ resulting in:

$$B_{eq} = \frac{1}{2\pi RC}\int_0^{+\infty} \frac{1}{1+F^2}\mathrm{d}F$$
$$= \frac{1}{2\pi RC}\left[\tan^{-1}(F)\right]_0^{+\infty} = \frac{1}{2\pi RC}\left[\frac{\pi}{2} - 0\right] = \frac{1}{4RC}$$

Part (b):

- $|H(f)| = 0$ only as $f \to \pm\infty$ implying that the null bandwidth is ∞.
- In this case, the null bandwidth definition does not give a useful measure of bandwidth because all RC low-pass filters will have $B_{null} = \infty$ regardless of the value of RC.

Part (c):

- As in part (b), since $|H(f)| = 0$ only as $f \to \pm\infty$ then the absolute bandwidth is ∞.

Problem 2-90:

- $\mathcal{P}_s(f)$ can be written as:

$$\mathcal{P}_s(f) = \mathrm{sinc}^2\left(f/B_n\right)$$

- The equivalent bandwidth is given by:

$$B_{eq} = \frac{1}{[\mathcal{P}_s(f)]_{f=0}}\int_0^{+\infty} \mathcal{P}_s(f)\mathrm{d}f = \int_0^{+\infty} \mathrm{sinc}^2\left(f/B_n\right)\mathrm{d}f$$

Since $\mathrm{sinc}^2\left(f/B_n\right)$ is an even function, then:

$$B_{eq} = \frac{1}{2}\int_{-\infty}^{+\infty} \mathrm{sinc}^2\left(f/B_n\right)\mathrm{d}f$$

- Since $\mathrm{sinc}\left(f/B_n\right)$ is the Fourier transform of $B_n\mathrm{rect}(B_nt)$, then applying Parseval's theorem results in:

$$B_{eq} = \frac{1}{2}\int_{-\infty}^{+\infty} \left[\mathrm{sinc}\left(f/B_n\right)\right]^2\mathrm{d}f$$
$$= \frac{1}{2}\int_{-\infty}^{+\infty} \left[B_n\mathrm{rect}(B_nt)\right]^2\mathrm{d}t$$
$$= \frac{1}{2}B_n^2\int_{-\frac{1}{2B_n}}^{+\frac{1}{2B_n}} 1.\mathrm{d}t = \frac{1}{2}B_n^2\frac{1}{B_n} = \frac{B_n}{2}$$

Problem 2-92:

- The signal and its Fourier transform are given by:

$$s(t) = \text{tri}\left(\frac{t}{T_0}\right) \quad \Rightarrow \quad S(f) = T_0 \text{sinc}^2(T_0 f)$$

Part (a):

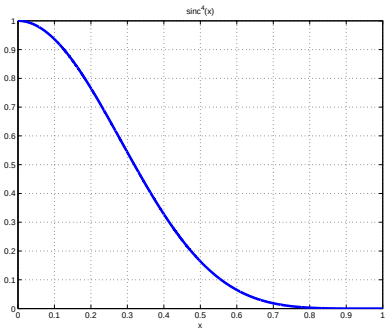
- The absolute bandwidth is infinity since $S(f)$ extends between $-\infty$ and $+\infty$.

Part (b):

- The 3-dB bandwidth B_{3dB} can be determined from:

$$\begin{aligned} |S(B_{3dB})|^2 &= \frac{1}{2}\left[\max|S(f)|^2\right] = \frac{1}{2}\left[|S(f)|^2\right]_{f=0} = \frac{1}{2}T_0^2 \\ \Rightarrow \quad & T_0^2 \text{sinc}^4(T_0 B_{3dB}) = \frac{1}{2}T_0^2 \\ \Rightarrow \quad & \text{sinc}^4(T_0 B_{3dB}) = \frac{1}{2} \end{aligned}$$

- Using a computer, the function $\text{sinc}^4(x)$ is plotted below for x between 0 and 1:



From this figure, we observe that $\text{sinc}^4(x) = 1/2$ for $x \simeq 0.319$ implying that:

$$B_{3dB} = \frac{0.319}{T_0}$$

Part (c):

- The equivalent bandwidth is given by:

$$\begin{aligned} B_{eq} &= \frac{1}{\left[|S(f)|^2\right]_{f=0}} \int_0^{+\infty} |S(f)|^2 \mathrm{d}f = \frac{1}{T_0^2} \int_0^{+\infty} T_0^2 \text{sinc}^4\left(T_0 f\right) \mathrm{d}f \\ &= \int_0^{+\infty} \text{sinc}^4\left(T_0 f\right) \mathrm{d}f \end{aligned}$$

given that $\text{sinc}^4\left(T_0 f\right)$ is an even function, then:

$$B_{eq} = \frac{1}{2}\int_{-\infty}^{+\infty} \text{sinc}^4\left(T_0 f\right) \mathrm{d}f$$

- Since $\text{sinc}^2(T_0 f)$ is the Fourier transform of $\frac{1}{T_0}\text{tri}\left(\frac{t}{T_0}\right)$, then applying Parseval's theorem results in:

$$
\begin{aligned}
B_{eq} &= \frac{1}{2}\int_{-\infty}^{+\infty}\left[\text{sinc}^2(T_0 f)\right]^2 \mathrm{d}f \\
&= \frac{1}{2}\int_{-\infty}^{+\infty}\left[\frac{1}{T_0}\text{tri}(t/T_0)\right]^2 \mathrm{d}t \\
&= \frac{1}{2}\frac{1}{T_0^2}\left[2\int_{0}^{+\infty}\text{tri}^2(t/T_0)\,\mathrm{d}t\right] = \frac{1}{T_0^2}\int_{0}^{T_0}\text{tri}^2(t/T_0)\,\mathrm{d}t \\
&= \frac{1}{T_0^2}\int_{0}^{T_0}\left(1-\frac{t}{T_0}\right)^2 \mathrm{d}t = \frac{1}{T_0^2}\int_{0}^{T_0}\left(1+\frac{t^2}{T_0^2}-2\frac{t}{T_0}\right)\mathrm{d}t \\
&= \frac{1}{T_0^2}\left[T_0+\frac{1}{T_0^2}\frac{1}{3}T_0^3-\frac{2}{T_0}\frac{T_0^2}{2}\right] \\
&= \frac{1}{T_0^2}T_0\left[1+\frac{1}{3}-1\right] = \frac{1}{3T_0}
\end{aligned}
$$

Part (d):

- $S(f) = T_0\text{sinc}^2(T_0 f)$ is equal to zero when $T_0 f$ is a nonzero integer implying that f is a nonzero integer multiple of $1/T_0$. Consequently, the first zero crossing is at $f = 1/T_0$ implying that:

$$B_{null} = \frac{1}{T_0}$$